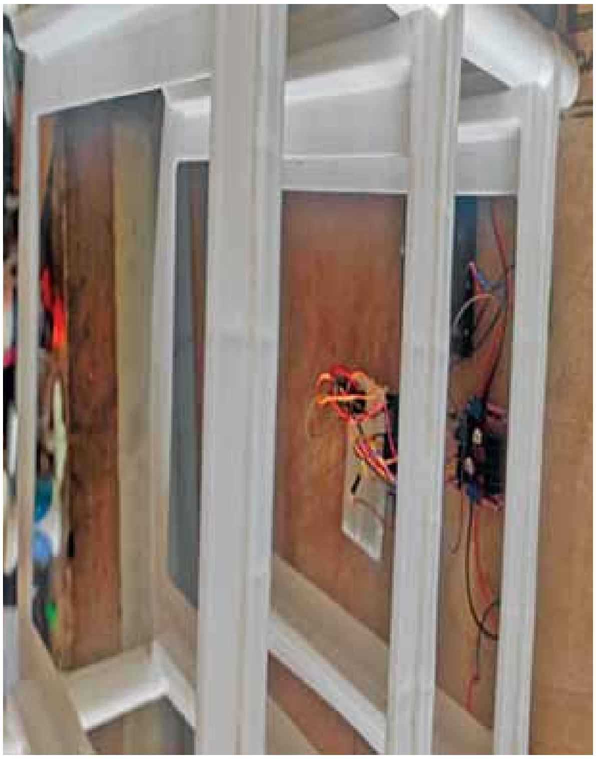

Fig. 4: Robot chassis

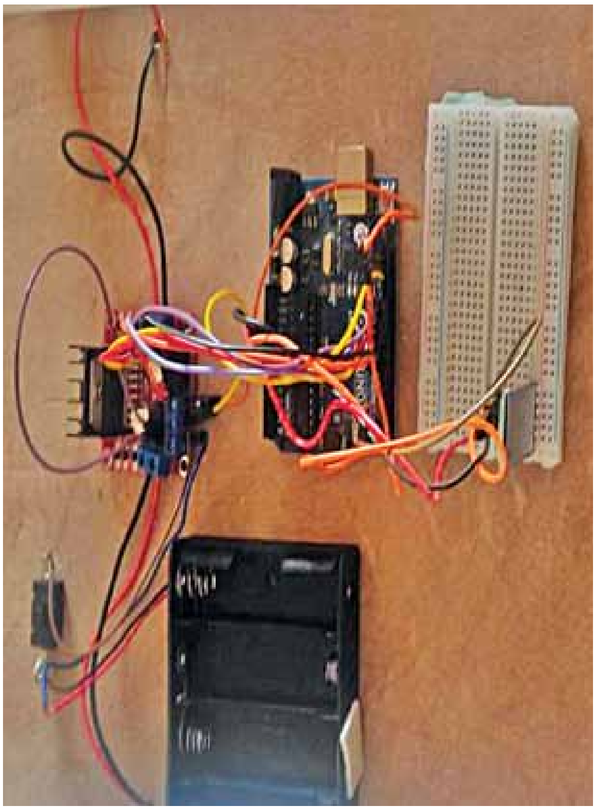

Fig. 5: Robot circuit

campuses, or events to serve water to staff during examinations or functions, reducing human effort, time, and cost.

All functions are written on the Arduino platform and saved in the sketch as Waterrobot.ino. Fig. 3 shows a snippet of the source code.

Before assembly, upload the source code to Arduino by selecting the proper board and port.

Construction and testing

To build this robot, Bluetooth technology was utilised along with electronic components, including Arduino Uno, Bluetooth module (HC-05), L298 driver, 12V, 500 RPM geared DC motor, 100cm diameter wheel with a 6mm shaft, breadboard, jumper wires, USB cable, wooden plank, battery holder, and battery module 18650 3.7V 2000mAh. The programming of the device is done on the Arduino software.

The L298N-based motor driver module 2A is a high-power motor driver perfect for driving DC geared motors. It can control up to 4 DC motors, or 2 DC motors with directional and speed control.

EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*

Make connections as per the circuit diagram. Provide voltage from the battery holder 18650 Li-ion cell. Check the status of the HC-05 Bluetooth sensor; if the LED is glowing, it is operational.

Once the HC-05 is on, open the serial Bluetooth terminal app on your mobile, switch on your Bluetooth, check the device list to connect, and connect with HC-05 from the list. You may set a password like '0000' or '1234' if prompted. Once the mobile is connected to HC-05, send signals like characters 'f' for forward, 'b' for backward, 'r' for right, 'l' for left, and 's' to stop to control the vehicle.

Mechanical assembly

Cut a wooden plank to make the serving robot chassis, then attach the wheels to the motors. On top of the chassis, affix a container for holding water or food to be served to people. Alternatively, a ready-made robot body with container can be purchased from the market.

Fig. 4 shows the robot chassis used by the author, while Fig. 5 illustrates how the assembled circuit on the breadboard is fixed onto the robot circuit. EFY

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